

A Universal Reliability Platform for GaN HEMTs with Realistic Stress Emulation, Online Monitoring, and In-Circuit Characterization

Dilip Rana, Tian Qiu, and Zheyu Zhang
Electrical, Computer and Systems Engineering
Rensselaer Polytechnic Institute, School of Engineering
Troy, New York, United States
ranad@rpi.edu, qiut2@rpi.edu, zhangz49@rpi.edu

Abstract—Realistic emulation of application operating conditions and effective degradation tracking are essential for accurate reliability assessment of power semiconductors. For gallium nitride (GaN) high electron mobility transistors (HEMTs), even minor modifications to the gate and power loops—introduced for online monitoring or offline characterization—can significantly alter stress dynamics due to the high sensitivity of circuit parasitics at fast switching speeds. This paper presents a universal reliability test platform for GaN HEMTs, applicable across a wide range of power electronic applications, that accurately emulates application-level stressors while supporting both online degradation monitoring and a novel offline in-circuit characterization approach. An overview of the test platform is first presented, covering its implementation and control strategy. The normal operation mode is then described alongside the real-time degradation tracking methodology. A circuit-based offline characterization method is subsequently presented, enabling extraction of I-V characteristics directly from the device within its operating circuit without hardware modifications. Finally, the platform is experimentally validated using commercial GaN HEMTs stressed in a synchronous buck converter configuration, confirming its accuracy and practicality for both online monitoring and offline characterization.

Index Terms—GaN, Realistic Stress Emulation, Online Monitoring, In-Circuit Characterization, I-V Characterization

I. INTRODUCTION

Wide bandgap (WBG) devices have established themselves as the future of power electronics due to their fast switching speeds, low power losses, and high breakdown voltages and ability to work under extreme temperatures for different applications ranging from Electric Vehicles, Motor Drivers, Renewables etc. [1]. While silicon carbide (SiC) devices are considered suitable for high-power applications, GaN devices are found best suited for high-frequency applications [2]. Among different GaN structures, only GaN high electron mobility transistor (HEMT)s are fully commercialized yet [3].

To understand the reliability of GaN HEMTs, two main approaches have been adopted: the failure-oriented approach and the application-oriented approach. The failure-oriented approach focuses on identifying and understanding specific failure mechanisms. Common tests under this category include passive thermal cycling [4], static gate stress [5], switching gate stress [6], step stress testing [7], hard-switching tests

[8], and backgating tests [9]. In contrast, the application-oriented approach aims to replicate and evaluate reliability under actual use-case conditions. A widely used commercial method for assessing reliability under such application stress is to operate devices until failure — a method known as test-to-fail. Examples include [10], [11], which used an open-buck circuit for high temperature operating life (HTOL) testing, and [12], which implemented an application board with power recycling. Similarly, converter efficiency was monitored during the dynamic high temperature operating life (DHTOL) test in [13], while resistive circuit HTOL testing was performed in [14].

However, degradation tracking through tests enables more comprehensive reliability modeling than simply waiting for device failure. One such approach is online monitoring. In this context, [15] proposed a boost-like topology to emulate and monitor stress under hard switching conditions only, while [16] introduced a full-bridge inductor topology to emulate both hard and soft switching conditions. [17] presented a three-phase back-to-back topology for AC power cycling tests with online turn-on resistance measurement, focusing on temperature stress by controlling electrical loss. Similarly, [18] developed an online monitoring method for GaN devices that adjusts for dynamic on-resistance offsets by adding an additional calibration mode before scanning. Likewise, [19] demonstrated online monitoring for power cycling tests on SiC MOSFET devices.

Another approach to degradation tracking is periodic offline characterization, which can provide more detailed insights into device degradation. For example, [20] performed offline I-V characterization using curve tracers after power cycling tests; however, this method is not easily adaptable for application-oriented reliability characterization. Similarly, [21] proposed a synchronous buck converter topology with offline measurement using curve tracers and additional relay terminals to switch connections to the measurement unit.

However, due to the fast switching speeds of GaN HEMTs, overshoots and transients are highly sensitive to circuit parasitics, as shown by [22]–[24]. This sensitivity has direct implications for reliability modeling; as demonstrated by [25],

even small gate transients can significantly impact device lifetime. Consequently, modifications made to these platforms for degradation tracking and offline monitoring can severely alter stress conditions, potentially leading to misleading reliability results.

This paper addresses the challenges of accurately evaluating the reliability of GaN HEMTs by emulating realistic stress conditions under actual operating scenarios and monitoring degradation. This is achieved by replicating the gate and power loop design of the target application circuit, keeping the circuit parasitics very close to the application circuit while enabling degradation monitoring through both online and offline characterization. First, an overview of the proposed test platform is provided, including its physical implementation and overall test strategy. Second, the platform's normal operation mode is described, along with the approach used for online degradation monitoring. Third, an offline characterization method is introduced to extract the I-V characteristics of GaN HEMTs, highlighting its potential applications beyond conventional reliability testing. Finally, the test platform is designed, implemented, and experimentally evaluated to validate its functionality and feasibility.

II. PLATFORM OVERVIEW

Fig. 1(a) illustrates the simplified circuit diagram of the proposed reliability test platform, which consists of three main components: a GaN board, a normal operation circuit, and an offline characterization circuit. The GaN board houses the device under test (DUT)s and connects either to the normal operation circuit, where the DUTs are subjected to realistic application-level stress conditions for continuous online degradation monitoring, or periodically to the offline characterization circuit, which is to extract detailed I-V characteristics for comprehensive degradation tracking. The platform's modular design enables seamless switching between these two modes without interfering with the operation of the other.

In the GaN board, the DUTs are arranged in a phase-leg configuration, along with gate drivers and decoupling

capacitors. The layout is carefully designed to closely replicate the gate and power loop of a real application circuit, ensuring accurate reproduction of switching oscillations and voltage overshoots. Each DUT is equipped with Kelvin connections for precise drain-source voltage measurement. The gate driver receives control signals from an external controller and operates according to the selected mode. A common connector is used to interface the GaN board with both the normal operation and offline characterization circuits, helping maintain circuit parasitics consistent with those of practical applications. Although a relay-based mechanism could be employed to automate switching between modes depending on parasitic tolerance, manual switching is preferred to minimize parasitics, especially given the focus on high-frequency operation.

The normal operation circuit is designed to replicate application-level stress conditions and enable online monitoring. It uses a power supply in conjunction with an electronic load to flexibly control electrical stressors as needed. In offline characterization mode, the I-V characteristics of the DUTs are extracted using a circuit-based setup rather than a commercial curve tracer. This approach not only offers a more cost-effective solution but also enables direct characterization of the DUTs within the actual operating circuit—an approach we refer to as in-circuit characterization.

Fig. 1(b) shows the hardware implementation of the setup configured for normal operation. As seen, the GaN board is placed inside an environmental chamber to replicate mission-specific or accelerated stress conditions, while the remaining circuit components are kept at room temperature to maintain system reliability, enabled by the modular design. Part of the normal operation circuit is directly attached to the chamber enclosure to minimize circuit parasitics. The rest of the equipment and circuits are placed outside the chamber and controlled by a central PC. For measurements, an oscilloscope is used and continuously interfaced with the control PC. Additionally, the temperature of the environmental chamber is managed by a temperature controller, which is also controlled via the central PC.

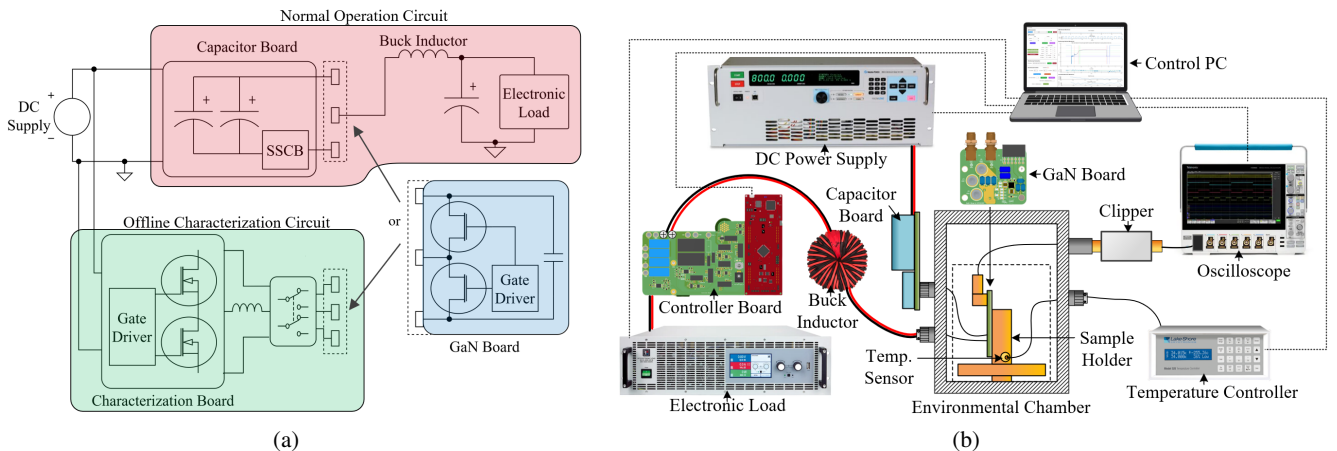


Fig. 1: Proposed test platform: (a) simplified circuit diagram; (b) hardware implementation configured for normal operation

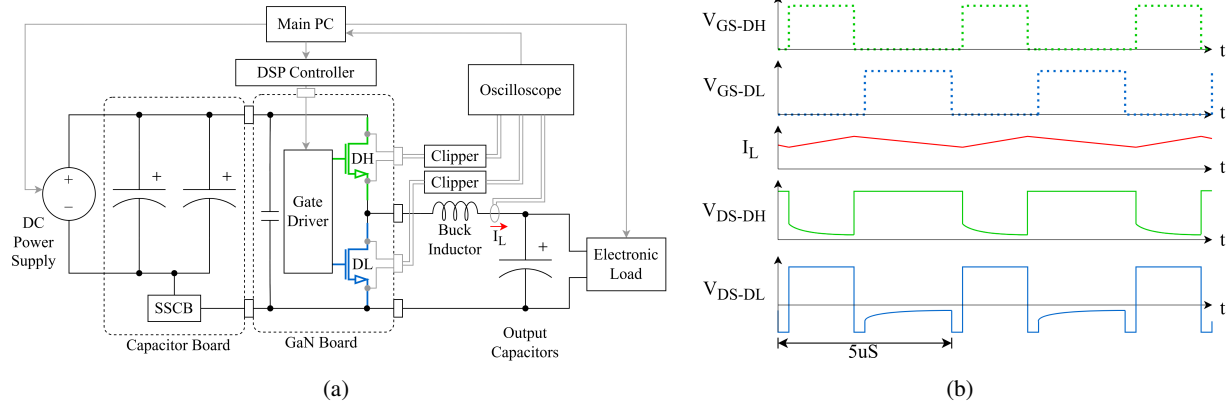


Fig. 2: Test platform under normal operation mode: (a) circuit diagram; (b) operation waveforms

III. NORMAL OPERATION MODE

In normal operation mode, the DUTs are continuously stressed using a stressor profile that follows the actual application mission profile. Alternatively, an accelerated test profile can be applied to shorten test time and reduce resource usage. Since the GaN board arranges the DUTs in a phase-leg configuration—a common setup in power electronics applications—the platform can replicate a wide range of application scenarios using one or more GaN boards. In this work, a single GaN board is used with a synchronous buck converter selected as the representative example.

Fig. 2(a) shows the complete circuit diagram of the test platform configured for normal operation mode. As illustrated, a DC power supply is connected to a capacitor board that includes energy storage capacitors and an solid-state circuit breaker (SSCB) for protection. Since the platform operates in normal mode for most of the testing duration, it must be not only reliable but also capable of protecting the equipment against unforeseen failures—hence the inclusion of the SSCB. The capacitor board is connected to the GaN board, which receives gate driver signals from an external digital signal processor (DSP) controller. The phase-leg output of the GaN board is connected to a buck inductor and output capacitor, forming a synchronous buck converter. An electronic load is connected at the output to serve as the load for the converter.

To apply the desired stressors to the DUTs, both electrical and environmental parameters are carefully controlled. Electrical stressors are managed by adjusting the voltage across the DUTs, the current flowing through them, and their switching events. The DC bus voltage is regulated using a programmable power supply, while the power handled by the DUTs—and consequently the current through them—is controlled using an electronic load, which offers flexible and precise control; this load can be replaced with a passive load for simplicity or depending on specific test requirements. Switching events are controlled through gate driver signals, allowing implementation of either a realistic application-level control strategy or a test-specific switching pattern. In this study, the gate control is configured to regulate the output voltage at a fixed

value, consistent with the intended application, to ensure accurate stress replication. Environmental stressors are controlled via the environmental chamber; for instance, a temperature-humidity chamber can be used to replicate mission-specific environmental conditions. As shown in the circuit diagram, all equipment—including the power supply, electronic load, DSP controller, and environmental chamber—is managed by a central PC, ensuring synchronization of parameters and coordinated operation.

For online monitoring, the drain-source voltages of the DUTs and the current through the buck inductor are measured in real time, as shown in the figure. Kelvin connections are used for accurate drain-source voltage measurements. However, because the drain-source voltage transitions rapidly between a high DC level and a very small on-state voltage, direct oscilloscope measurements at the Kelvin points can introduce errors. To mitigate this, voltage clippers are used to clip the DC bus voltage level, thereby improving the accuracy of on-state voltage measurements. The oscilloscope is continuously interfaced with the control system to capture DUT parameters and track degradation.

The waveforms during normal operation mode are shown in Fig. 2(b). As seen, the DUTs operate in complementary PWM mode with a small dead time for synchronous buck converter operation. The inductor current remains nearly constant, exhibiting only a small switching ripple. The drain-source voltage of the upper DUT displays a relaxation-type behavior due to the dynamic on-resistance characteristics of GaN HEMTs when turned on, and is clipped when the device is turned off. A similar trend is observed for the lower DUT, along with additional behavior from freewheeling conduction. While monitoring the voltage during freewheeling conduction may provide further insight into device degradation, the clipper must be capable of handling negative voltages at the expected magnitude. To track on-resistance degradation, including its dynamic behavior, both the inductor current and drain-source voltage are used. Since the inductor current flows only through the fully-on DUT, it represents the drain current. Together with the measured drain-source voltage, this enables the calculation

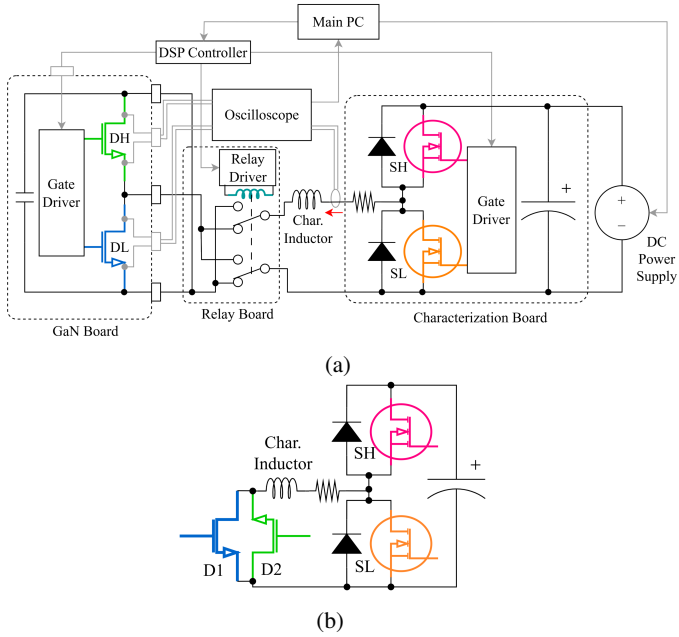


Fig. 3: Circuit diagram of the test platform under offline characterization mode: (a) complete circuit; (b) simplified circuit

of on-resistance and its dynamic variations, which can be used to monitor DUT degradation throughout the test.

IV. OFFLINE CHARACTERIZATION MODE

The complete circuit diagram of the test platform in the offline characterization mode is shown in Fig. 3(a). As illustrated, the GaN board connects to a relay board housing a double-pole double-throw (DPDT) relay and its driver, with control signals provided by the DSP controller. The opposite side of the relay board connects to a characterization inductor, which then connects to a characterization board. This characterization board features a phase leg equipped with energy storage capacitors, gate drivers, and a resistor at the phase leg output. This setup forms an antiparallel connection of the DUTs in series with the inductor and resistor at the output of the characterization phase leg, as shown in Fig. 3(b).

The characterization circuit operates by applying a ramp current through the DUTs to obtain their I-V characteristics. The characterization phase leg is controlled to apply a voltage pulse, which charges the inductor and generates a controlled current ramp through the DUTs, followed by inductor discharge. A series resistor is included to limit the maximum current and dissipate the inductor's energy during discharge. The relay board allows switching the DUTs between positions D_1 and D_2 , enabling characterization in both positive and negative conduction modes. By appropriately controlling the gate signals, the DUTs can be turned on and off to capture I-V characteristics during both turn-on and freewheeling conduction.

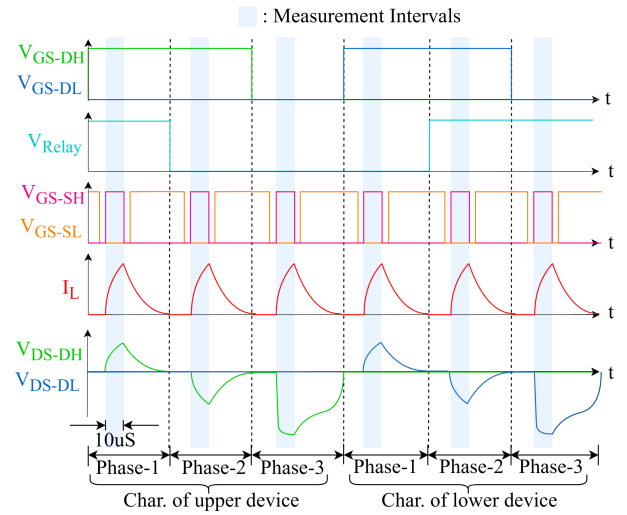


Fig. 4: Operation waveforms for offline characterization

The waveform corresponding to the characterization process is shown in Fig. 4. As illustrated, both DUTs are characterized sequentially, with each characterization cycle divided into three distinct phases, targeting the DUTs under positive conduction, negative conduction, and freewheeling conduction, respectively. In all phases, the relay is operated first and given sufficient time to fully engage before activating the gate of the DUTs and operating the characterization phase leg, to account for relay switching delays. To mitigate the current collapse effect commonly observed in GaN HEMTs, the gate of the DUT being characterized is turned on in advance before the application of the ramp current. In Phase 1, the relay positions the DUT being characterized at D_1 , and upon applying the gate signal, the characterization phase leg is operated, allowing a ramp current to flow from drain to source through the DUT being characterized. The drain-source voltage and inductor current are monitored during this interval, and since the current flows exclusively through the DUT being characterized, its I-V characteristics in positive conduction mode can be accurately extracted. After this, the phase leg is used to discharge the inductor current before moving to the next phase. In Phase 2, the relay repositions the DUT being characterized to D_2 , enabling current to flow from source to drain; the same gate and phase leg control sequence is used to obtain I-V characteristics in negative conduction mode. Finally, in Phase 3, the configuration remains as in Phase 2, but the gate signal is removed, allowing characterization of the DUT in freewheeling conduction mode.

Since only three short ramp pulses are sufficient to obtain the I-V characteristics of a DUT, the proposed method enables rapid, near-instantaneous characterization compared to traditional curve tracer-based approaches, which typically apply multiple rectangular pulses at discrete levels to construct the I-V curve. This not only reduces the total characterization time but also minimizes potential degradation of the DUT during the process. Moreover, using ramp currents instead of

rectangular pulses significantly reduces the influence of circuit parasitics on measurement accuracy. Although the current can be made to rise very slowly, the rise rate must be limited to avoid excessively long pulse durations and to minimize self-heating effects in the DUT. The setup allows both DUTs in the phase leg to be characterized sequentially and almost instantaneously without any changes to the hardware connections. Additionally, each segment of the I-V characteristic can be independently extracted, as the characterization phases and their execution order can be flexibly controlled as needed.

This in-circuit characterization setup enables practical diagnostics beyond traditional reliability testing, supporting application-level use cases such as health monitoring, degradation tracking, and troubleshooting of power converters during system downtime. By simply incorporating Kelvin connections to the DUTs, devices configured in a phase-leg topology can be characterized almost instantaneously and with high accuracy—an approach not achievable using conventional characterization methods. Furthermore, I-V characteristics at various gate voltages can be obtained by implementing a tunable gate drive, depending on diagnostic requirements. However, in most cases, characterization at the operational gate voltage is of primary interest and can typically be achieved without any additional circuit modifications.

V. EXPERIMENTAL VALIDATION

To validate both the functionality and feasibility of the reliability test platform across its operating modes, it was physically implemented and experimentally tested. Commercially available EPC-2307 GaN HEMTs (200 V, 62 A, 10 mΩ) were used as DUTs on the GaN board. During normal operation, a DC bus voltage of 70 V and a switching frequency of 200 kHz were applied, with the power stage designed to handle up to 1 kW of output power. Current and voltage measurements were performed using a 6-channel oscilloscope, with CLP1500V15A1 clippers employed for accurate drain-source voltage monitoring. For offline characterization, the setup was optimized to apply a 10 μs ramp current pulse to cover the required characterization range.

The hardware setup for experimental validation is shown in Fig. 5. Various components of the system are visible, including the circuit elements, DC power supply, electronic load, and an auxiliary power supply used to power the gate drivers and control circuitry. In the figure, the controller board—which integrates both the DSP controller and the characterization circuitry into a single PCB—is directly mounted on the enclosure of the environmental chamber. The GaN board is interfaced through connectors on the environmental chamber and is mounted on a heatsink located inside the chamber. For clarity, the GaN board is shown outside the enclosure in the figure to highlight relevant parts of the system.

In Fig. 6(a), the steady-state on-resistance of a DUT, captured from online degradation monitoring over approximately 90 hours, is shown. Additional filtering was applied to reduce measurement noise, revealing almost no degradation during

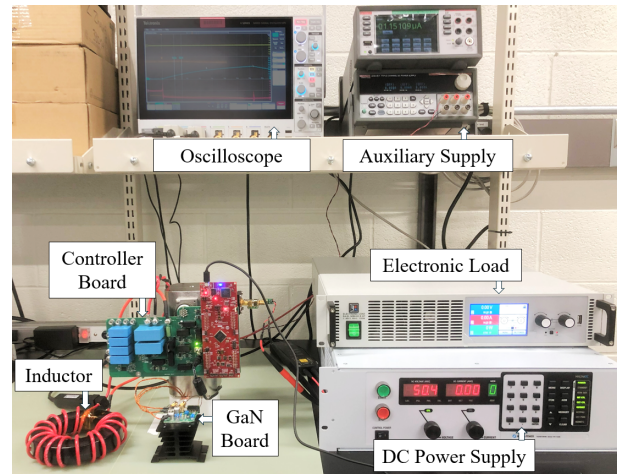


Fig. 5: Hardware setup for experimental validation

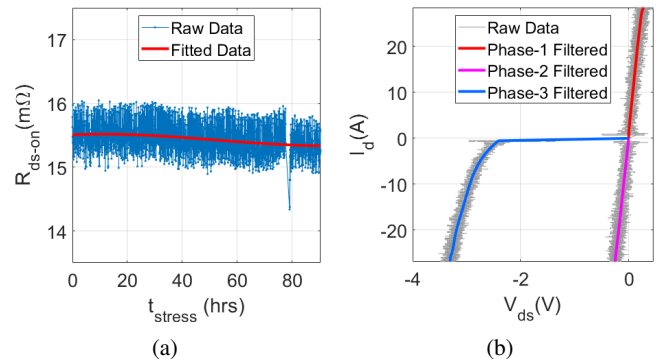


Fig. 6: Platform test result from: (a) normal operation mode; (b) offline characterization mode

this period. This validates the platform’s capability for accurate online degradation monitoring.

Fig. 6(b) presents the I-V characteristics of a DUT obtained from the offline in-circuit characterization setup at room temperature, capturing positive conduction, negative conduction, and freewheeling conduction behaviors. Although minor noise was present, it was effectively filtered, resulting in curves that closely match the datasheet specifications. In the linear region for both conduction directions, an on-resistance of approximately 10 mΩ was extracted, demonstrating the method’s effectiveness in mitigating the current collapse effect. Additionally, the freewheeling conduction exhibited knee-voltage-like behavior consistent with the body diode-like characteristics of the GaN HEMT. These results validate the platform’s capability to perform precise and reliable offline in-circuit characterization.

VI. CONCLUSION

This paper presents a universal reliability platform that emulates realistic stress conditions under application scenarios and supports both online monitoring and offline in-circuit characterization for degradation tracking. By preserving

the gate and power loop design of the actual application, the platform replicates realistic electrical stress environments with high fidelity. Experimental results demonstrate that on-resistance degradation can be effectively tracked throughout testing by simply incorporating Kelvin connections across the DUTs. An offline, circuit-based characterization setup was also implemented, allowing I-V characterization without requiring hardware modifications, and delivering results with satisfactory accuracy. In addition to being cost-effective, the setup enables direct offline characterization of DUTs within their operating circuits, extending its use beyond reliability testing to include practical diagnostics such as health monitoring and troubleshooting in real-world power converter applications.

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